



600 HOURS SKILL ACQUISITION

Study DevOps with Linux, AWS, Kubernetes & Terraform

Linux Commands Understanding Based Exercise compiled

by Engr. Kah Kissinger

Initial Setup (Everyone starts here):

1. **Task:** Verify your current working directory.
 - **Expected Output:** Your current working directory should be your home directory (e.g., /home/yourusername or similar).
-

Exercise Steps:

Part 1: Directory Navigation & Creation

1. **Task:** Create a new directory named my_linux_project in your current home directory.
 - **Show Output:** Show the contents of your home directory to confirm the new directory exists.
2. **Task:** Navigate into the my_linux_project directory.
 - **Show Output:** Show your current working directory.
3. **Task:** Inside my_linux_project, create three new subdirectories named documents, scripts, and data.
 - **Show Output:** List the contents of my_linux_project to show the new subdirectories.
4. **Task:** Navigate into the documents directory.
 - **Show Output:** Show your current working directory.
5. **Task:** Go back up one level to the my_linux_project directory.
 - **Show Output:** Show your current working directory.
6. **Task:** Navigate directly from my_linux_project to the scripts directory using an absolute path (assuming your home directory is /home/yourusername).
 - **Show Output:** Show your current working directory.

7. **Task:** Navigate directly to your home directory from scripts without using `cd ..` multiple times.
 - **Show Output:** Show your current working directory.
8. **Task:** Go back to the `my_linux_project` directory.
 - **Show Output:** Show your current working directory.

Part 2: File Creation & Manipulation

9. **Task:** Inside `my_linux_project`, create an empty file named `README.md`.
 - **Show Output:** List the contents of `my_linux_project` to show the new file.
10. **Task:** Create a file named `notes.txt` inside the `documents` directory. Add the text "My important notes for the project." to this file.
 - **Show Output:** Display the content of `notes.txt`.
11. **Task:** Create a file named `hello.sh` inside the `scripts` directory. Add the text `#!/bin/bash` on the first line and `echo "Hello, Linux!"` on the second line.
 - **Show Output:** Display the content of `hello.sh`.
12. **Task:** Copy the `README.md` file from `my_linux_project` into the `documents` directory.
 - **Show Output:** List the contents of the `documents` directory to show the copied file.
13. **Task:** Rename the `README.md` file *within the documents directory* to `project_info.md`.
 - **Show Output:** List the contents of the `documents` directory to show the renamed file.
14. **Task:** Move the `project_info.md` file from `documents` back to the `my_linux_project` directory.
 - **Show Output:** List the contents of `my_linux_project` to show the moved file, and then list the contents of `documents` to show it's no longer there.
15. **Task:** Create a new empty file named `temp_file.txt` in your current working directory (which should be `my_linux_project`).
 - **Show Output:** List the contents of `my_linux_project` to show the new file.
16. **Task:** Delete the `temp_file.txt` file.
 - **Show Output:** List the contents of `my_linux_project` to show the file is gone.

Part 3: Advanced Navigation & Permissions (Conceptual)

17. **Task:** Try to navigate into a directory you know exists on your system but is *not* in your home directory (e.g., /etc). You may not have permissions to modify it, but you should be able to enter it.
 - **Show Output:** Show your current working directory after navigating to the specified location.
18. **Task:** Navigate back to your my_linux_project directory using the shortest possible command from /etc.
 - **Show Output:** Show your current working directory.
19. **Task:** Make the hello.sh script in the scripts directory executable.
 - **Show Output:** Show the long listing (ls -l) of the scripts directory to demonstrate the permission change for hello.sh.
20. **Task:** Attempt to execute the hello.sh script from the my_linux_project directory (you'll need to specify its path).
 - **Show Output:** Show the output of the script execution.
21. **Task:** Create a new directory named backup inside my_linux_project.
 - **Show Output:** List the contents of my_linux_project to show the new directory.
22. **Task:** Copy the entire documents directory and its contents into the backup directory.
 - **Show Output:** List the contents of the backup directory to show the copied documents directory within it, and then list the contents of backup/documents to confirm its files are there.
23. **Task:** Delete the original documents directory and all its contents (be careful!).
 - **Show Output:** List the contents of my_linux_project to show documents is gone.
24. **Task:** Create a symbolic link named my_notes_link in my_linux_project that points to the notes.txt file which is now inside backup/documents.
 - **Show Output:** Show a long listing (ls -l) of my_linux_project to display the symbolic link and where it points.
25. **Task:** Use the symbolic link my_notes_link to display the content of notes.txt.
 - **Show Output:** Display the content of notes.txt through the symbolic link.

Compiled by Engr. Kah Kissinger